# Financial Analytics & Risk Reporting System

## 1. Introduction

The Financial Analytics & Risk Reporting System is an end-to-end analytics project focused on understanding lending operations, financial performance, and loan default risk.

The project combines PostgreSQL, Power BI, Python, and machine learning to explore financial data and generate analytical reports.

The datasets used in this project are synthetic and were generated for learning and demonstration purposes.

## 2. Project Objectives

The main objectives of this project are:

- Generate and prepare financial and lending datasets.
- Analyze loan and financial information using PostgreSQL.
- Build interactive dashboards in Power BI.
- Explore borrower and loan characteristics associated with default.
- Develop machine learning models to predict loan default status.
- Save and reuse a trained model for future predictions.

## 3. Technologies Used

| Technology | Purpose |
|---|---|
| Python | Data generation, preprocessing, and machine learning |
| Pandas | Data manipulation and cleaning |
| NumPy | Numerical operations |
| PostgreSQL | Data storage and SQL analysis |
| Power BI | Dashboard development and reporting |
| Scikit-learn | Machine learning algorithms |
| Joblib | Saving and loading the trained model |
| VS Code | Development environment |

## 4. Dataset Description

The project uses synthetic lending data representing a financial organization.

The datasets include information related to:

- Loans
- Customers
- Loan payments
- General ledger transactions
- Loan default status

The loan dataset contains 3,000 records and 15 columns.

Important loan attributes include:

| Column | Description |
|---|---|
| LoanID | Unique loan identifier |
| CustomerID | Customer identifier |
| LoanType | Type of loan |
| IssueDate | Date the loan was issued |
| LoanAmount | Amount of the loan |
| InterestRate | Loan interest rate |
| TermMonths | Loan duration in months |
| CreditScore | Borrower's credit score |
| DebtToIncomeRatio | Borrower's debt-to-income ratio |
| PriorDelinquencies | Previous delinquencies |
| AnnualIncome | Borrower's annual income |
| DefaultProbability | Generated default probability |
| DefaultFlag | Target variable |
| LoanStatus | Loan status |

## 5. Data Preparation

The dataset was loaded into Python using Pandas.

The following checks and preprocessing steps were performed:

- Checked dataset shape and column names.
- Reviewed data types.
- Checked for missing values.
- Checked for duplicate records.
- Checked duplicate loan identifiers.
- Examined the distribution of the target variable.

The dataset contained no missing values or duplicate records in the checks performed.

The target variable distribution was:

| DefaultFlag | Records | Percentage |
|---|---|---|
| 0 — Non-default | 2,041 | 68.03% |
| 1 — Default | 959 | 31.97% |

## 6. SQL Analysis

PostgreSQL was used to store and analyze the financial datasets.

SQL analysis was performed to explore loan and financial information and support reporting.

The SQL work formed part of the broader financial analytics workflow before dashboard development.

## 7. Power BI Dashboard

Power BI was used to create interactive financial and loan portfolio dashboards.

The dashboard supports visual exploration of financial data through KPIs, charts, and filters.

The purpose of the dashboard was to make financial and lending information easier to understand and analyze.

# 8. Machine Learning

## 8.1 Problem Definition

The machine learning task was to classify loans as default or non-default.

The target variable was:

DefaultFlag

- 0 — Non-default
- 1 — Default

This is a binary classification problem.

## 8.2 Feature Selection

The following columns were excluded:

- LoanID
- CustomerID
- DefaultProbability
- DefaultFlag
- LoanStatus

Loan identifiers were removed because they do not represent meaningful predictive information.

DefaultFlag was excluded from the input features because it is the target.

LoanStatus and DefaultProbability were excluded to avoid potential target leakage.

## 8.3 Feature Engineering

Date columns were converted into datetime format.

The following features were created:

- IssueYear
- IssueMonth
- LoanDurationDays

The original date columns were then removed.

### 8.4 Categorical Encoding

The LoanType column was categorical.

One-hot encoding was applied using Pandas get_dummies().

The resulting features included:

- LoanType_Auto
- LoanType_Business
- LoanType_Education
- LoanType_Home
- LoanType_Personal

After preprocessing, the dataset contained 15 numerical features.

### 8.5 Train-Test Split

The dataset was divided into training and testing sets.

| Dataset | Records |
|---|---|
| Training set | 2,400 |
| Testing set | 600 |

An 80:20 split was used.

The split used:

- random_state=42
- stratify=y

Stratification helped preserve the target class distribution in both sets.

## 9. Machine Learning Algorithms

Five classification algorithms were tested:

1. Decision Tree
2. Random Forest
3. Logistic Regression
4. Gradient Boosting
5. K-Nearest Neighbors

The algorithms were evaluated using the same train-test split.

**Model Accuracy Results**

| Algorithm | Accuracy |
|---|---|
| Decision Tree | 69.17% |
| Random Forest | 76% |
| Logistic Regression | 76% |
| Gradient Boosting | 75% |
| K-Nearest Neighbors | 61% |

Random Forest was selected as the preliminary model based on the recorded accuracy results.

Random Forest and Logistic Regression had the same recorded accuracy, so accuracy alone does not establish that Random Forest is the superior model.

## 10. Model Evaluation

The Decision Tree model was evaluated using:

- Accuracy
- Confusion matrix
- Precision
- Recall
- F1-score

The Decision Tree accuracy was approximately 69.17%.

The confusion matrix was:

| | Predicted Non-default | Predicted Default |
|---|---|---|
| Actual Non-default | 315 | 93 |
| Actual Default | 92 | 100 |

The model correctly identified 100 actual defaulters and missed 92 actual defaulters.

This demonstrates why accuracy alone is not sufficient for evaluating a loan default prediction model.

## 11. Model Saving

The trained Random Forest model was saved using Joblib.

```
import joblib

joblib.dump(rf_model, "loan_default_model.pkl")
```

The saved model can be loaded without retraining.

```
model = joblib.load("loan_default_model.pkl")
```

## 12. Prediction Script

A separate Python file named prediction.py was created to load the trained model and predict the default status of a sample loan.

The prediction script:

1. Loads the saved model.
2. Creates sample loan input data.
3. Passes the input to the model.
4. Predicts the loan default status.
5. Displays the prediction.

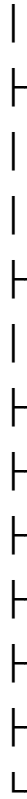
The sample prediction produced:

Prediction: No Loan Default

This was a demonstration using sample values.

## 13. Project Structure

Financial_Analytics_Project/



├── generated_data/

│ └── output/



├── validation_reports/



├── dataset_generation.py

├── loan_default_prediction.py

├── prediction.py

├── loan_default_model.pkl

├── requirements.txt

├── README.md

└── PROJECT_DOCUMENTATION.md

## 14. How to Run the Project

**Install dependencies**

pip install -r requirements.txt

**Train the model**

python loan_default_prediction.py

This runs preprocessing, trains the model, evaluates it, and saves the trained model.

**Run prediction**

python prediction.py

This loads the saved model and predicts the default status of the sample loan.

## 15. Limitations

- The project uses synthetic data rather than real financial records.
- Model performance may not represent actual lending behavior.
- Accuracy alone is insufficient to establish model quality.
- Further model tuning and evaluation are required.
- The model should not be used for real lending or financial decisions.

## 16. Conclusion

This project provided practical experience in financial data analysis, SQL, dashboard development, feature engineering, and machine learning.

It brought together data preparation, database analysis, visualization, model comparison, and prediction in a single project workflow.

The project demonstrates how analytics and machine learning can be applied to explore financial data and lending risk.

## Author

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